Retail Sales Dataset
Data Analysis Portfolio Project
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Date: **October 2026**
Dataset Source: **Kaggle Dataset**

Executive Summary

This project analyzes retail sales transactions to understand customer purchasing behavior, sales performance, product preferences, and pricing patterns. Using Excel Pivot Tables, Pivot Charts, and an interactive dashboard, the analysis identified the most valuable customer segments, top-performing product categories, monthly sales trends, and pricing insights. The findings demonstrate how data analysis can support inventory planning, marketing strategies, and customer segmentation decisions.

Introduction

Retail businesses rely on customer and sales data to make informed decisions regarding inventory management, product offerings, and marketing campaigns. This project explores a retail sales dataset containing transaction-level information, including customer demographics, product categories, sales amounts, and pricing details. The objective is to identify meaningful patterns and provide actionable business insights through exploratory data analysis (EDA).

Skills Demonstrated

Throughout this project, the following analytical and technical skills were applied:

- Data Cleaning and Validation
- Exploratory Data Analysis (EDA)
- Pivot Table Analysis
- Pivot Chart Visualization
- Dashboard Design and Development
- Data Storytelling
- Business Insight Generation
- Customer Segmentation Analysis
- Sales Trend Analysis
- Excel Slicers and Interactive Reporting

Dataset Profile
The dataset contains retail transaction records and customer demographic information. Key variables include:

- Transaction ID
- Date
- Customer ID
- Gender
- Age
- Product Category
- Quantity
- Price per Unit
- Total Amount

The dataset consists of approximately 1,000 retail transactions covering the period from January 2023 to December 2023 and includes three product categories: Beauty, Clothing, and Electronics.

Dataset Summary
Before analysis, the dataset was reviewed to ensure consistency and completeness. Customer demographic information, transaction details, product categories, and sales values were examined for accuracy. Data was then organized into structured tables to support Pivot Table analysis and dashboard creation.

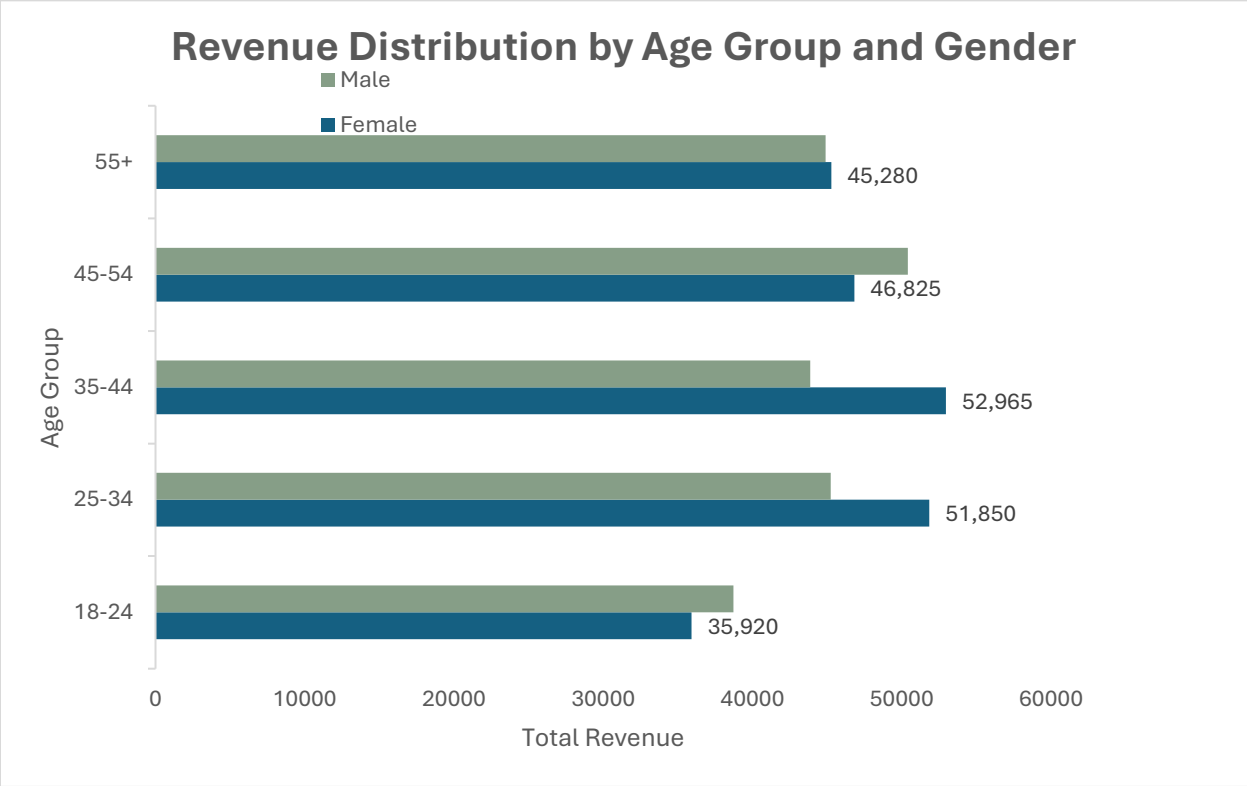
Purpose of Data Cleaning
Data cleaning is a critical step in the analytical process because accurate results depend on high-quality data. The cleaning process was performed to:

- Ensure date values were correctly recognized
- Verify category names were consistent
- Remove formatting inconsistencies
- Validate numerical fields such as Quantity, Price per Unit, and Total Amount
- Prepare the dataset for Pivot Table and dashboard analysis

These steps improved the reliability of the analysis and visualizations.

Analysis Questions
1. How do customer age and gender influence their purchasing behavior?
2. Are there discernible patterns in sales across different time periods?
3. Which product categories hold the highest appeal among customers?
4. What are the relationships between age, spending, and product preferences?
5. What insights can be gleaned from the distribution of product prices within each category?

Analysis 1



The relationship between customer age and gender was analyzed to determine how demographic factors influence spending behavior. Total revenue was aggregated across age groups and gender segments to identify high-value customer groups and compare purchasing patterns between male and female customers.

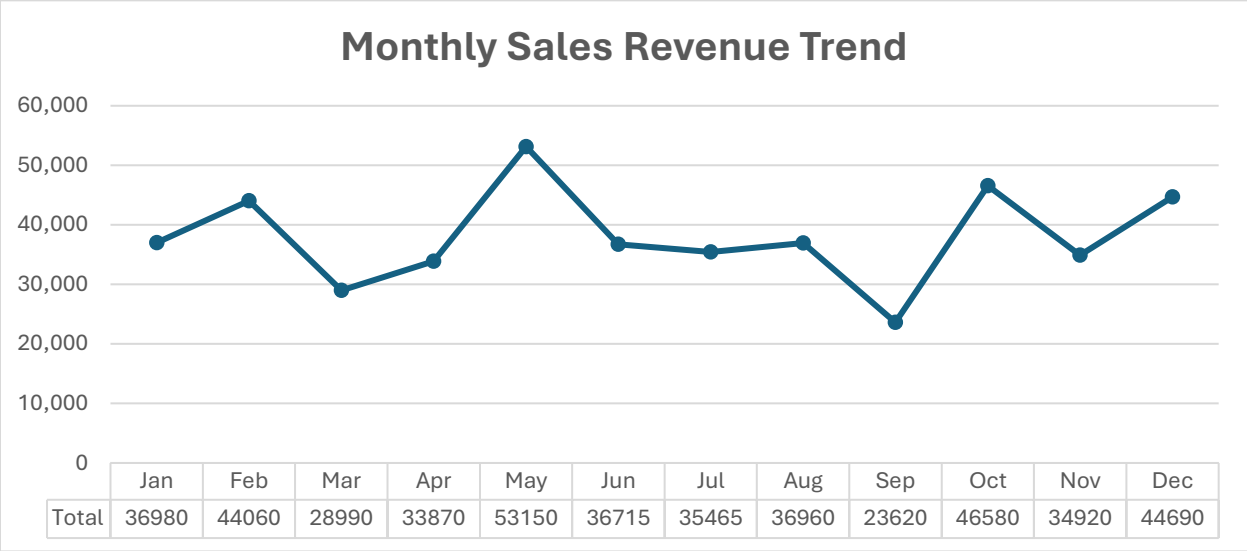
Key Finding

Customer purchasing behavior is influenced more by age than gender. The highest revenue comes from customers aged 45-54 (97,235), followed closely by the 25-34 and 35-44 age groups. This suggests that middle-aged customers are the retailer's most valuable customer segment. While female customers generated slightly higher overall revenue (232,840) than male customers (223,160), spending patterns remain relatively balanced across genders.

Business Insight

The findings suggest that age is a stronger driver of purchasing behavior than gender. Marketing efforts should focus on customers aged 25-54, as these groups contribute the largest share of revenue and represent the retailer's most valuable customer segments.

Analysis 2



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Total	36980	44060	28990	33870	53150	36715	35465	36960	23620	46580	34920	44690

Monthly sales revenue was analyzed to identify trends and fluctuations in customer purchasing activity throughout the year. Examining monthly performance helps reveal periods of high and low demand that may influence inventory planning and marketing strategies.

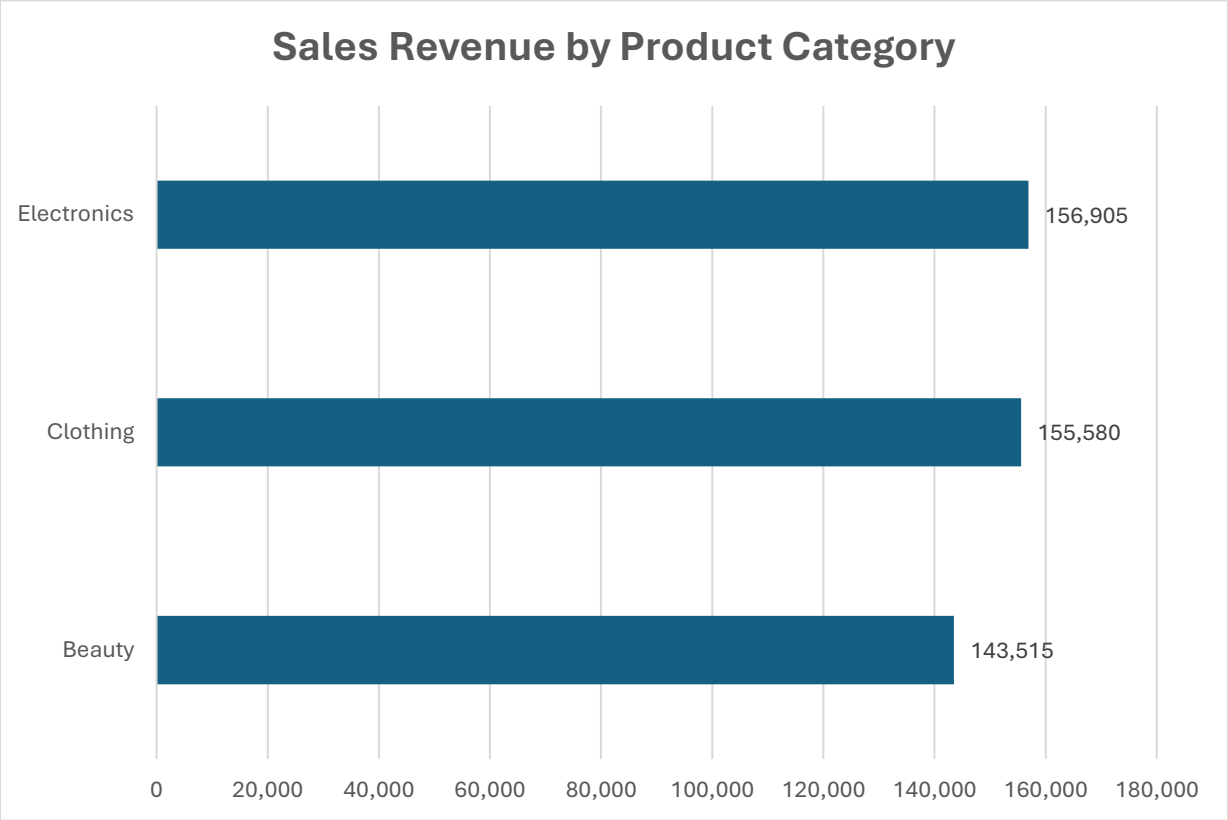
Key Finding

Monthly sales revenue fluctuated throughout the year, with the highest revenue recorded in May (53,150) and the lowest in September (23,620). Following the September decline, revenue recovered strongly in October (46,580) and remained relatively high through December (44,690), suggesting possible seasonal variations in customer demand.

Business Insight

Understanding monthly sales trends helps the business identify peak and low-performing periods, enabling more effective inventory planning, resource allocation, and promotional planning.

Analysis 3



Product category performance was evaluated by comparing total revenue generated across Beauty, Clothing, and Electronics categories. This analysis helps identify customer preferences and the categories that contribute most significantly to overall sales performance.

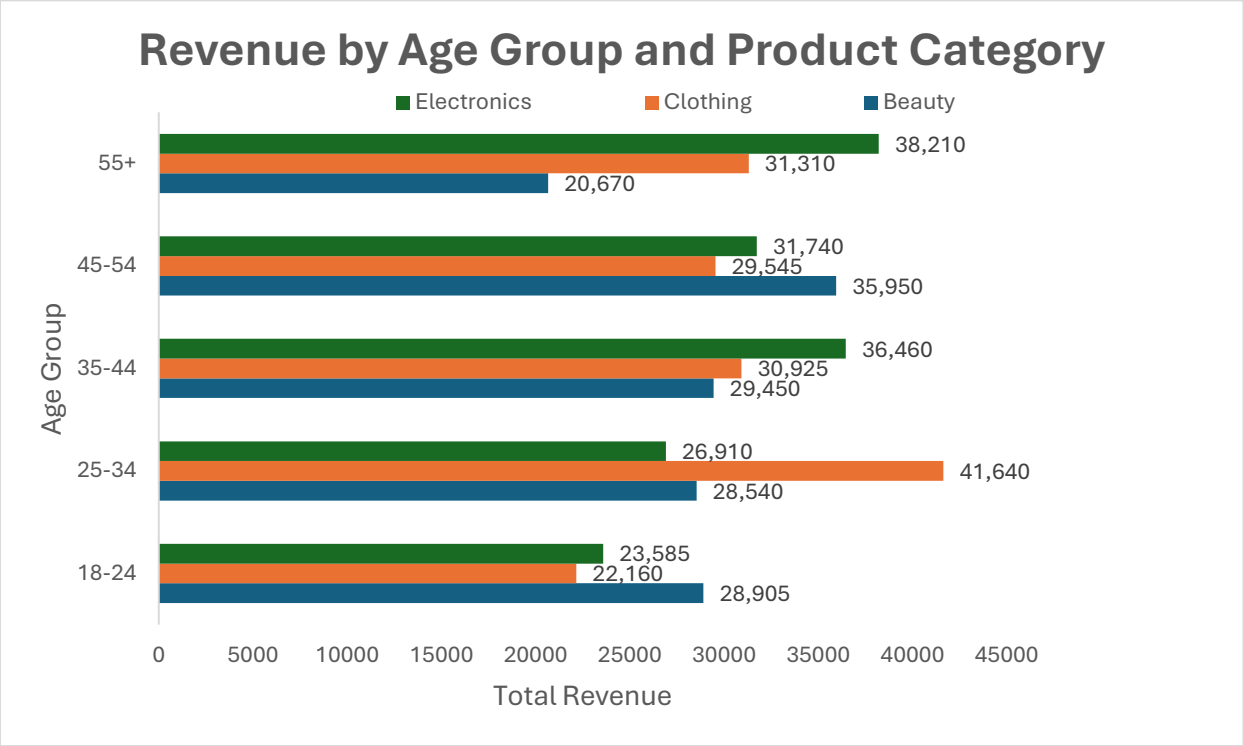
Key Finding

Product category analysis shows that all three categories contributed significantly to total revenue. Electronics generated the highest revenue (156,905), followed closely by Clothing (155,580), while Beauty recorded the lowest revenue (143,515). The relatively small differences between categories indicate balanced customer demand across the product range.

Business Insight

Electronics and Clothing are the strongest revenue-generating categories and should remain a key focus for inventory planning and promotional campaigns. Increasing customer engagement within the Beauty category may provide opportunities for future revenue growth.

Analysis 4



Revenue was analyzed by both age group and product category to understand how customer demographics influence purchasing preferences. This analysis provides deeper insight into the relationship between customer age, spending behavior, and product selection.

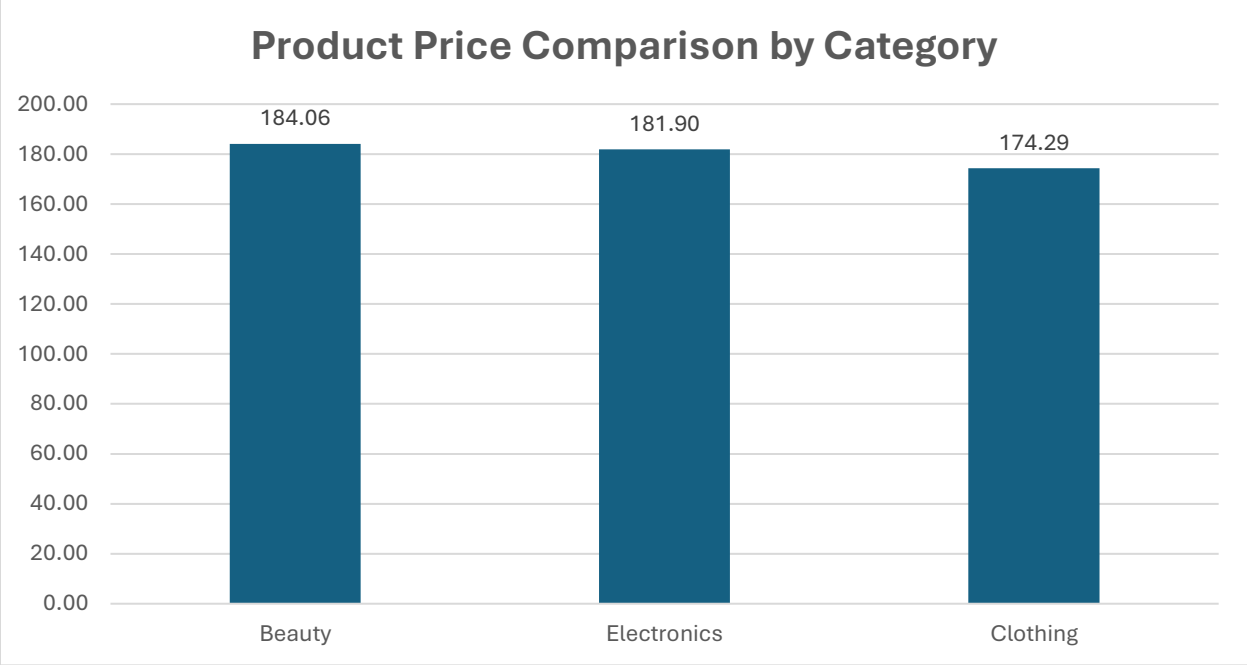
Key Finding

Customer spending and product preferences vary across age groups. While Electronics is the preferred category among customers aged 35-44 and 55+, Clothing generates the highest revenue for customers aged 25-34. Beauty products perform strongest among customers aged 18-24 and 45-54. These differences suggest that purchasing preferences change across different stages of life.

Business Insight

The retailer can improve marketing effectiveness by targeting age groups with products that align with their purchasing preferences. Age-based customer segmentation can support more personalized promotions and inventory planning.

Analysis 5



Average product pricing was compared across categories to determine pricing differences and evaluate whether higher prices correspond with stronger sales performance. This analysis helps assess the relationship between product pricing and customer demand.

Key Findings

The average price per unit was compared across product categories to evaluate pricing differences. Beauty products recorded the highest average price per unit (184.06), followed closely by Electronics (181.90). Clothing had the lowest average price per unit (174.29). The relatively small differences suggest a consistent pricing strategy across all categories.

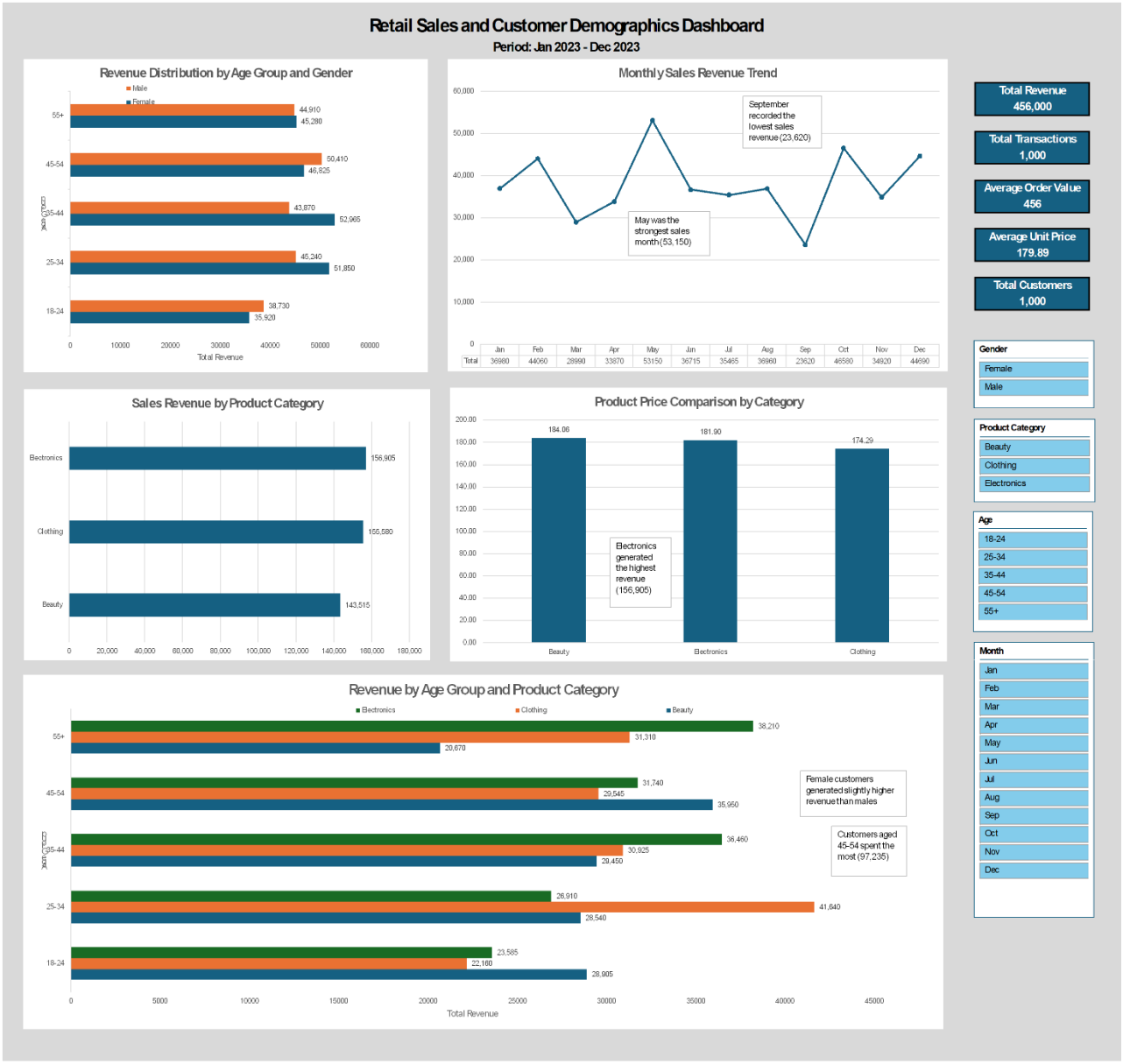
The findings suggest that product popularity and customer demand contribute more to sales success than price differences. Electronics combines strong customer demand with relatively high pricing, making it the retailer's strongest performing category.

Business Insight

Pricing alone does not determine sales success. Although Beauty products have the highest average price, Electronics generated the most revenue. This suggests that customer demand and product preferences have a greater influence on sales performance than pricing differences.

Dashboard

Interactive Excel Dashboard



Recommendations

1. Focus marketing efforts on customers aged 25-54, as they contribute the highest revenue.
2. Maintain strong inventory availability for Electronics and Clothing products due to their strong sales performance.
3. Investigate factors contributing to lower sales in September and implement promotional strategies during slower periods.
4. Utilize age-based customer segmentation to deliver more targeted marketing campaigns.
5. Monitor pricing strategies while prioritizing customer demand and product preferences to drive revenue growth.

Conclusion

The analysis successfully identified several important customer and sales trends within the retail dataset. Age was found to have a stronger influence on purchasing behavior than gender, with customers aged 25-54 contributing the highest sales revenue. Electronics emerged as the highest-performing product category, while customer demand varied across different age groups and product types. Monthly revenue trends revealed fluctuations throughout the year, with May recording the highest sales and September the lowest. Additionally, pricing analysis showed that Electronics achieved the highest revenue despite not having the highest average price, highlighting the impact of customer demand on sales performance.

Overall, the findings demonstrate how data analysis can support business decision-making through customer segmentation, product performance monitoring, and sales trend analysis. The interactive dashboard provides a user-friendly tool for exploring these insights and can assist management in developing targeted marketing, inventory, and pricing strategies.